A STEINER QUADRUPLE SYSTEM OF ORDER 38 WITH MINIMUM-COLORABLE DERIVED DESIGNS

Abstract. We construct a Steiner quadruple system of order 38 whose derived Steiner triple system at every point has block-chromatic index 19. Since every STS(37) requires at least 19 partial parallel classes, this is an mcDSQS(38), resolving the smallest unknown mcDSQS($6n+2$) case posed by Tan and Zhou. It also establishes the $n = 38$ case of Conjecture 2 of Shi, Xia, and Krotov and yields $q_0'(3, 4, 38) = 20$. The system is specified by 27 orbits under an affine group of order 111, and the two colorings that the proof needs are printed in full.

1. Result

A block coloring of an STS(v) is a partition of its blocks into partial parallel classes. For an STS(v) $\mathcal{D}$, let $\chi'(\mathcal{D})$ be the least number of classes in such a coloring, and set

$$\chi'(v) = \min\{\chi'(\mathcal{D}) : \mathcal{D} \text{ is an STS}(v)\}.$$

Following Tan and Zhou [3], an SQS(v) is an mcDSQS(v) if every derived STS($v-1$) has chromatic index $\chi'(v-1)$.

Theorem 1. *There exists an* mcDSQS(38)*. More precisely, there is an* SQS(38) *for which every derived* STS(37) *has block-chromatic index* 19*.*

Tan and Zhou asked for an mcDSQS(38) and identified it as the smallest unknown mcDSQS($6n+2$) [3, Sec. 4, item (1)]. Theorem 1 answers that part of their question; it does not address the RDSQS(52) requested in the same item. It also establishes the $n = 38$ instance of [2, Conj. 2]. Rotational SQS(38)s were already known: Feng, Chang, and Ji constructed rotational SQS($p+1$)s with a nontrivial multiplier for primes $p \equiv 13 \pmod{24}$ [1]. The value $\chi'(37) = 19$ is known as well: Tan and Zhou record $\chi'(v) = (v+1)/2$ for $v \equiv 1 \pmod 6$ with $v \notin \{7, 13\}$ [3]. The contribution here is the simultaneous minimum colorability of all 38 derived systems.

2. Construction

Let

$$X = \mathbb{Z}_{37} \cup \{\infty\}.$$

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All arithmetic on finite points is modulo 37. Since 10 has order 3 in $\mathbb{Z}_{37}^{\times}$, the maps

$$g_{a,b}(x) = 10^a x + b, \qquad a \in \{0,1,2\}, \quad b \in \mathbb{Z}_{37},$$

fixing ∞, form a group G of order 111. For $Q \subseteq X$, write $\mathrm{Orb}_G(Q) = \{g(Q) : g \in G\}$.

Consider the representatives

$$\mathcal{A} = \left\{ \begin{array}{lll} \{\infty,0,1,11\}, & \{\infty,0,2,22\}, & \{\infty,0,4,7\}, \\ \{\infty,0,8,14\}, & \{\infty,0,16,28\}, & \{\infty,0,19,32\} \end{array} \right\},$$

$$\mathcal{S} = \left\{ \begin{array}{lll} \{0,1,27,34\}, & \{0,2,17,31\}, & \{0,3,7,28\}, \\ \{0,2,19,24\}, & \{0,5,23,29\}, & \{0,1,12,28\} \end{array} \right\},$$

and

$$\mathcal{L} = \left\{ \begin{array}{lll} \{0,1,13,14\}, & \{0,1,8,21\}, & \{0,1,9,20\}, \\ \{0,1,10,31\}, & \{0,1,16,23\}, & \{0,1,2,7\}, \\ \{0,1,5,33\}, & \{0,1,4,18\}, & \{0,1,17,30\}, \\ \{0,1,3,15\}, & \{0,1,19,35\}, & \{0,2,9,15\}, \\ \{0,2,8,33\}, & \{0,2,18,23\}, & \{0,2,25,32\} \end{array} \right\}.$$

Define

$$\mathcal{B} = \bigcup_{Q \in \mathcal{A} \cup \mathcal{S} \cup \mathcal{L}} \mathrm{Orb}_G(Q).$$

Proposition 2. *The pair $(X, \mathcal{B})$ is an* SQS(38)*.*

Proof. If $g_{a,b}$ fixes a block with k finite points of sum σ, then summing those points gives $kb = (1 - 10^a)\sigma$ in $\mathbb{Z}_{37}$, so b is determined by a; as $|G| = 111$, each stabilizer therefore has order 1 or 3. Testing the two candidates for each representative shows that each representative in $\mathcal{A} \cup \mathcal{S}$ has stabilizer of order 3, generated by $g_{1,b}$ with

$$b = 1,2,4,8,16,32 \ \text{ on } \mathcal{A}, \qquad b = 27,17,7,19,29,28 \ \text{ on } \mathcal{S}$$

in the order listed, and that each representative in $\mathcal{L}$ has trivial stabilizer. A direct computation confirms that the 27 orbits are pairwise disjoint. Thus

$$|\mathcal{B}| = 12 \cdot 37 + 15 \cdot 111 = 2109.$$

Enumerating the triples contained in these blocks gives

$$\binom{38}{3} = 8436$$

distinct triples, each with multiplicity one. Hence every triple of X lies in a unique block of $\mathcal{B}$. $\square$

For $x \in X$, let

$$\mathcal{D}_x = \big(X \setminus \{x\}, \{B \setminus \{x\} : x \in B \in \mathcal{B}\}\big)$$

be the design derived at x. Proposition 2 implies that $\mathcal{D}_x$ is an STS(37) for every $x \in X$.

3. Minimum colorability

Lemma 3. *Every* STS(37) *has block-chromatic index at least* 19*.*

Proof. An STS(37) has $37 \cdot 36/6 = 222$ blocks, whereas a partial parallel class contains at most $\lfloor 37/3 \rfloor = 12$ blocks. Therefore every block coloring has at least

$$\left\lceil \frac{222}{12} \right\rceil = 19$$

classes. □

The action of G has the two point orbits $\{\infty\}$ and $\mathbb{Z}_{37}$. The multiplier $\mu = g_{1,0}$ fixes ∞ and 0, so it is an automorphism of both $\mathcal{D}_\infty$ and $\mathcal{D}_0$. Tables 1 and 2 list, for each of these two designs, a set C_0 of ten blocks and six sets $G_1, \ldots, G_6$ of twelve blocks each; the classes are C_0 together with the eighteen sets

$$\mu^j(G_i) \setminus C_0 \qquad (1 \le i \le 6,\ 0 \le j \le 2).$$

This partitions each of $\mathcal{D}_\infty$ and $\mathcal{D}_0$ into 19 partial parallel classes, in both cases with class-size profile

$$12^{14}, 11^4, 10;$$

that is, fourteen classes of size 12, four of size 11, and one of size 10. Every listed triple is a block of the relevant derived design, the triples within each class are pairwise disjoint, and the classes partition all 222 blocks.

Table 1. A 19-coloring of $\mathcal{D}_\infty$, whose point set is $\mathbb{Z}_{37}$.

C_0	$\{1,10,26\}$ $\{2,15,20\}$ $\{3,11,17\}$ $\{5,13,19\}$ $\{6,18,27\}$ $\{7,33,34\}$
	$\{8,16,22\}$ $\{9,35,36\}$ $\{14,29,31\}$ $\{21,25,28\}$
G_1	$\{0,30,34\}$ $\{2,8,31\}$ $\{3,11,17\}$ $\{4,5,15\}$ $\{6,19,24\}$ $\{7,9,29\}$
	$\{10,12,32\}$ $\{13,21,27\}$ $\{14,22,28\}$ $\{16,20,23\}$ $\{18,33,35\}$ $\{25,26,36\}$
G_2	$\{0,5,24\}$ $\{1,17,29\}$ $\{3,8,27\}$ $\{4,16,25\}$ $\{6,10,13\}$ $\{7,12,31\}$
	$\{9,35,36\}$ $\{11,23,32\}$ $\{14,18,21\}$ $\{15,19,22\}$ $\{20,28,34\}$ $\{26,30,33\}$
G_3	$\{0,23,31\}$ $\{1,6,25\}$ $\{2,28,29\}$ $\{3,26,34\}$ $\{4,19,21\}$ $\{7,13,36\}$
	$\{8,16,22\}$ $\{9,10,20\}$ $\{11,15,18\}$ $\{12,24,33\}$ $\{14,27,32\}$ $\{17,30,35\}$
G_4	$\{0,1,11\}$ $\{2,18,30\}$ $\{3,16,21\}$ $\{4,6,26\}$ $\{5,31,32\}$ $\{8,20,29\}$
	$\{9,22,27\}$ $\{10,14,17\}$ $\{12,13,23\}$ $\{15,28,33\}$ $\{19,34,36\}$ $\{24,25,35\}$
G_5	$\{0,16,28\}$ $\{1,24,32\}$ $\{2,7,26\}$ $\{3,29,30\}$ $\{4,12,18\}$ $\{5,20,22\}$
	$\{6,21,23\}$ $\{8,17,33\}$ $\{9,11,31\}$ $\{10,19,35\}$ $\{13,25,34\}$ $\{15,27,36\}$
G_6	$\{0,2,22\}$ $\{1,9,15\}$ $\{3,18,20\}$ $\{4,8,11\}$ $\{5,14,30\}$ $\{6,7,17\}$
	$\{10,25,27\}$ $\{12,16,19\}$ $\{13,26,31\}$ $\{23,24,34\}$ $\{28,32,35\}$ $\{29,33,36\}$

For $p \in \mathbb{Z}_{37}$, translation by p belongs to G and sends $\mathcal{D}_0$ to $\mathcal{D}_p$. It therefore transports the coloring at 0 to every finite point.

Proof of Theorem 1. Proposition 2 gives an SQS(38). The two tables and translation give a 19-coloring of every derived design, while Lemma 3 rules out fewer than 19 colors. Hence

$$\chi'(\mathcal{D}_x) = 19 \qquad (x \in X).$$

TABLE 2. A 19-coloring of $\mathcal{D}_0$, whose point set is $(\mathbb{Z}_{37} \setminus \{0\}) \cup \{\infty\}$.

C_0	$\{1,5,33\}$ $\{3,4,30\}$ $\{6,8,23\}$ $\{7,19,26\}$ $\{9,12,16\}$ $\{10,13,34\}$ $\{11,27,36\}$ $\{14,21,29\}$ $\{17,22,35\}$ $\{18,24,32\}$
G_1	$\{1,16,23\}$ $\{2,14,36\}$ $\{4,6,12\}$ $\{5,18,20\}$ $\{7,19,26\}$ $\{8,13,31\}$ $\{9,22,28\}$ $\{10,24,35\}$ $\{11,21,25\}$ $\{15,27,33\}$ $\{17,29,32\}$ $\{30,34,\infty\}$
G_2	$\{1,2,7\}$ $\{3,5,14\}$ $\{4,20,26\}$ $\{8,18,27\}$ $\{9,10,21\}$ $\{11,24,33\}$ $\{12,34,35\}$ $\{13,19,28\}$ $\{15,17,\infty\}$ $\{16,29,36\}$ $\{22,23,25\}$ $\{30,31,32\}$
G_3	$\{1,10,31\}$ $\{2,8,33\}$ $\{3,6,22\}$ $\{4,16,27\}$ $\{5,35,36\}$ $\{7,18,25\}$ $\{11,13,23\}$ $\{12,15,20\}$ $\{14,21,29\}$ $\{17,28,34\}$ $\{19,32,\infty\}$ $\{24,26,30\}$
G_4	$\{1,24,25\}$ $\{2,6,13\}$ $\{3,19,34\}$ $\{4,21,23\}$ $\{5,10,17\}$ $\{7,12,14\}$ $\{8,20,32\}$ $\{9,18,31\}$ $\{15,29,35\}$ $\{16,28,\infty\}$ $\{22,27,30\}$ $\{26,33,36\}$
G_5	$\{1,9,20\}$ $\{2,27,35\}$ $\{3,13,16\}$ $\{4,10,22\}$ $\{5,12,32\}$ $\{6,29,\infty\}$ $\{7,8,24\}$ $\{11,18,30\}$ $\{14,33,34\}$ $\{15,19,25\}$ $\{21,31,36\}$ $\{23,26,28\}$
G_6	$\{1,11,\infty\}$ $\{2,3,21\}$ $\{4,25,34\}$ $\{5,26,31\}$ $\{6,7,16\}$ $\{8,22,29\}$ $\{9,32,35\}$ $\{10,23,27\}$ $\{12,13,36\}$ $\{14,15,30\}$ $\{17,19,33\}$ $\{20,24,28\}$

The construction also shows $\chi'(37) \leq 19$, and Lemma 3 gives the reverse inequality. Thus $\chi'(37) = 19$, so every $\mathcal{D}_x$ is minimum colorable. $\square$

Corollary 4. *Conjecture 2 of Shi, Xia, and Krotov holds for* $n = 38$*, and*

$$q_0'(3,4,38) = 20.$$

Proof. For an STS, two vertices of the minimum-distance graph of Shi, Xia, and Krotov are adjacent exactly when the corresponding triples meet. Since two distinct Steiner triples meet in at most one point, the independent sets of this graph are precisely partial parallel classes. Their chromatic number is therefore the block-chromatic index used here. By [2, Thm. 5],

$$q_0'(3,4,38) = 1 + \min_{\mathcal{S} \text{ an SQS}(38)} \max_{\mathcal{D} \in \mathcal{S}'} \chi'(\mathcal{D}),$$

where $\mathcal{S}'$ is the set of derived systems of $\mathcal{S}$. Lemma 3 makes the quantity after 1 at least 19, and Theorem 1 attains 19. The same theorem is exactly the $n = 38$ assertion of [2, Conj. 2]. $\square$

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